SMART INDIA HACKATHON 2026

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Problem Statement ID	SIH 26160
Problem Statement Title	AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework
Theme	Blockchain & Cybersecurity
PS Category	Software
Team ID	SIH 163
Team Name	Highlanders

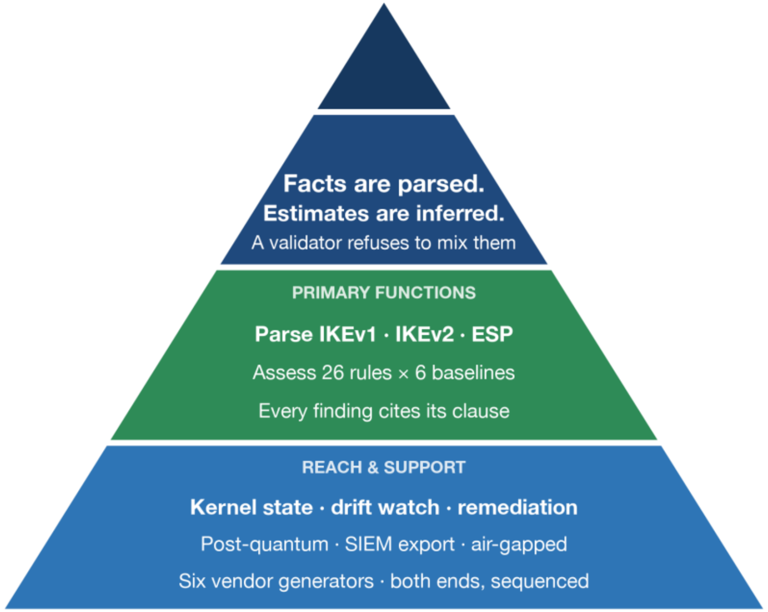
PROPOSED SOLUTION

One capture in — graded, cited evidence out

THE PROBLEM
Wireshark decodes an IKE exchange. It has no opinion about it, so weak IPsec survives for years.

THE IDEA
26 deterministic rules judge what was negotiated, each citing the clause it applies.

WHAT IS NEW
Facts are parsed, estimates inferred — and a validator refuses to mix them.



Depth of capability, from one principle

GAP		WHAT WE BUILT
Wireshark shows, never judges	→	26 cited rules, clause by clause
Rekey is encrypted — tunnel goes dark	→	Kernel + daemon state read and graded
Payload is encrypted — what does it carry?	→	Flow-shape inference, calibrated, abstains
A one-sided fix is an outage	→	Both ends, sequenced, zero packet loss

Two boundaries the design will not cross: it never writes to a network device, and stores no credential — an AST sweep over every module enforces both.

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TECHNICAL APPROACH

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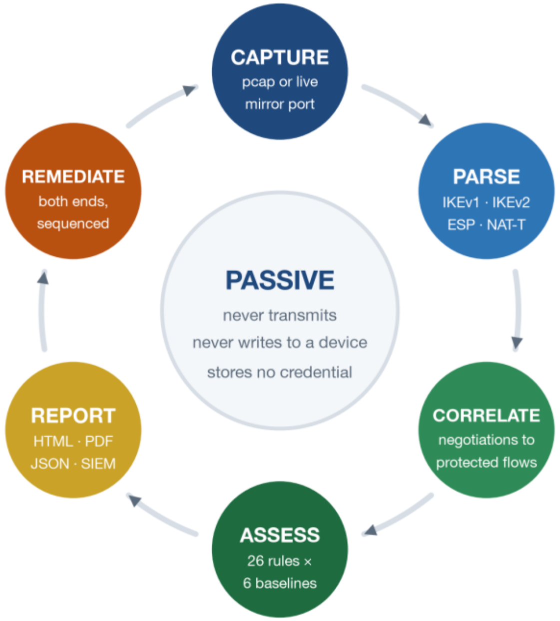
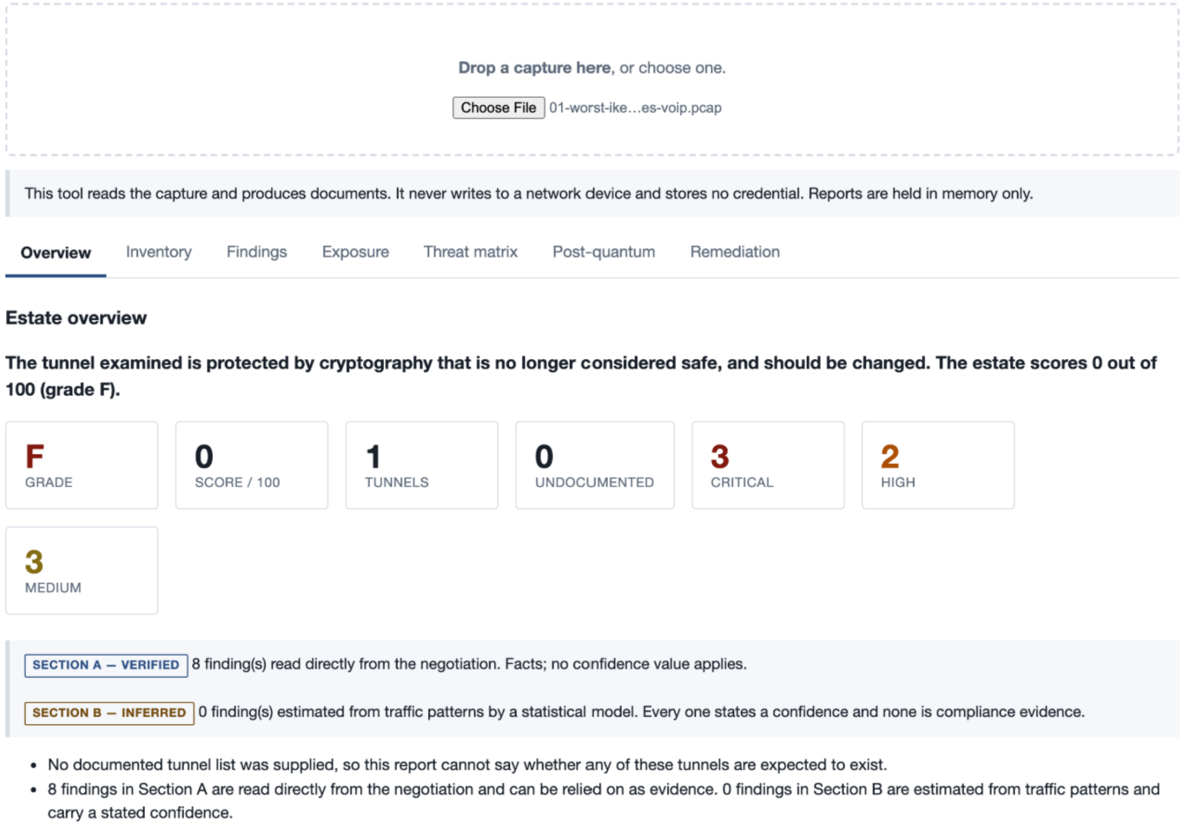
How it works, and what it is built from





Working prototype — the dashboard on a real estate capture
One tunnel, Graded F — eight findings, every finding cited.

1. **PARSE** — scapy-free streaming reader
2. **MODEL** — pydantic v2 · strict schemas
3. **LEARN** — scikit-learn · SHAP
4. **SERVE** — FastAPI · Jinja2 · WeasyPrint
5. **PROVE** — pytest · mypy --strict · ruff
6. **TESTBED** — strongSwan · Docker · netem

IKEv1 · IKEv2 · ESP | **26 rules × 6 baselines** | **6 vendor generators** | **NAT-T 4500 · both registries**

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FEASIBILITY AND VIABILITY

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What makes it work, what could go wrong

FEASIBILITY

- Working prototype, v1.0.0 tagged
- Runs on a laptop: 0.68 GB container
- No agent, no credential, no outage

VIABILITY

- Reads captures operators already have
- Air-gapped install, offline bundle
- ITSAR + CERT-In: no rival ships these

CHALLENGES & RISKS

- Model trained on laboratory traffic
- 5 of 6 vendor configs not device-tested
- A capture can miss an encrypted rekey

STRATEGIES

- Publish held-out numbers, worst included
- Every generated file states its own status
- Read kernel + daemon state to close it

Every performance target beaten

Margin over the required limit, measured on the bundled captures

2 613	306	93.8%	20/20
unit tests	integration, live pairs	coverage	M11 acceptance

Deployable today
0.68 GB container · non-root · runs under --network none · offline bundle

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IMPACT AND BENEFITS

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Who it helps, and what it changes

WHO **BENEFITS** and how

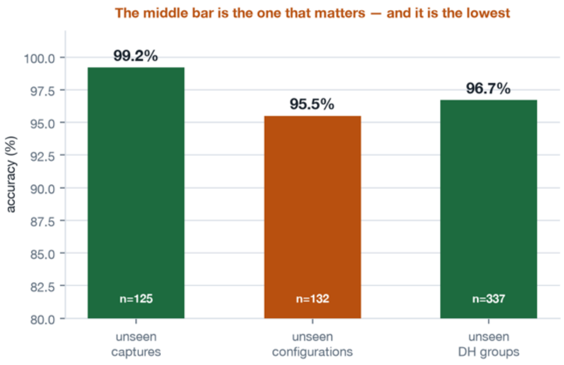
Auditors
cited evidence, not an opaque score

Network teams
both-ends fix, zero packet loss

Regulators
ITSAR & CERT-In from one capture

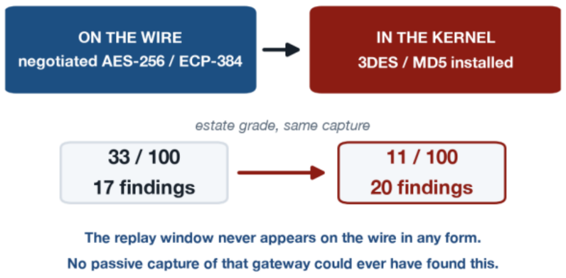
Air-gapped sites
runs offline, stores no credential

1. **Verifiable**
every finding cites its clause
2. **Fast**
100 MB capture assessed in 0.5 s
3. **Honest**
abstains when the signal is thin
4. **Future-proof**
post-quantum readiness graded



Honest about its limits

637 rows, 7 classes of laboratory traffic. The unflattering number is published, not the best one.



Endpoint visibility

Kernel state parsed and graded against the same rules — never fetched, never with a credential.

Social and economic benefit

Government, telecom and enterprise VPN estates get audit-grade evidence from a capture they already have — no agent, no credential, no outage, and an Indian baseline (ITSAR, CERT-In) that no comparable tool ships.

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RESEARCH AND REFERENCES

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The standards it applies, and the data it learned from

RFC 7296	RFC 2408 2409	RFC 8221 8247	RFC 4303	RFC 9370 8784	NIST 800-77 800-131A	ITSAR CERT-In
IKEv2	ISAKMP · IKEv1	algorithm reqs	ESP · anti-replay	post-quantum	baselines	India

Every one of the 26 rules cites the clause it applies — docs/RULES.md is generated from the registry

DATASETS — named in the problem statement

CIC-IDS2017 · CSE-CIC-IDS2018 · UNSW-NB15 · CTU-13 · CICIoT2023 · LANL · DARPA

Audited packet by packet, no sampling: none contains IKE, ESP or AH, so none carries a crypto label. Used as replayed inner payload — real traffic shape, our configuration supplies the labels.

DATASETS — generated for this project

strongSwan testbed corpus — 637 flow records · 7 classes · 36 configurations

Labels read back from what the tunnel actually negotiated. Every cipher meets at least two DH groups, so the classifier cannot learn the cipher instead of the traffic. Also used: ISCXVPN2016, MAWI, MITRE ATT&CK, NVD CVE.

Code: github.com/PrashamJ17/SIH_PS160 — v1.0.0 · M11 acceptance 20/20 · published JSON report schema 1.3

Evidence and demo: Video Demo